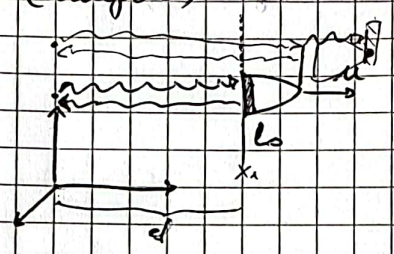
CORREZIONE ESAME SCRITTO 24/07/2023

1) CINEMATICA

(semplice)



$$\Delta t = t_B - t_A$$

$$t_A = \cancel{\frac{2d}{c}}$$

$$t_B = \frac{d}{c} + 2t^* + \frac{d}{c} = \cancel{\frac{2d}{c}} + 2t^*$$

→ il Δt è il tempo che il segnale impiega a "percorrere l'astronave" mentre questa si muove

$$ct^* = \frac{L_0}{\gamma} + vt^* \rightarrow t^* = \frac{L_0}{\gamma(c-v)}$$

$$\Delta t = \frac{2L_0}{\gamma c(1-\beta)} = \frac{2L_0}{c}\frac{\sqrt{1-\beta^2}}{1-\beta} = \frac{2L_0}{c}\sqrt{\frac{1+\beta}{1-\beta}} \rightarrow$$

$$\rightarrow \frac{1+\beta}{1-\beta} = \frac{c^2\Delta t^2}{4L_0^2} = 3 \rightarrow \beta = 0{,}5$$

NON utilizzare il tempo proprio/dilatato

$x_1' = 0 \qquad x_2' = L_0$

$t_1' = 0 \qquad t_2' = \frac{L_0}{c}$

$$t = \gamma\left(t' + \frac{v}{c^2}x'\right) = \gamma\left(\frac{L_0}{c} + \frac{v}{c^2}L_0\right)$$

$$\Delta t = 2\gamma\frac{L_0}{c}(1+\beta) = \frac{2L_0}{c}\sqrt{\frac{1+\beta}{1-\beta}}$$

2) DOPPLER

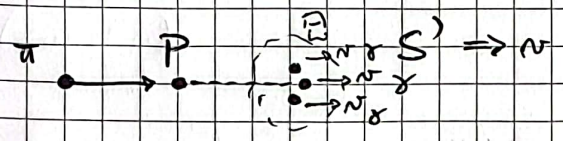
$$\lambda = \lambda_0 \sqrt{\frac{1+\beta}{1-\beta}} \longrightarrow \frac{1-\beta}{1+\beta} = \left(\frac{\lambda}{\lambda_0}\right)^2 = 3 \longrightarrow \beta = 0{,}5$$

$$\gamma = 1{,}155$$

$$\boxed{\alpha = \gamma^3 a_x}$$

3) DINAMICA

STANDARD, ci sarà anche ai prossimi appelli



$$\underbrace{E^2 - p^2c^2} = \underbrace{E'^2 - \cancel{p'^2c^2}} \rightarrow$$

$$\rightarrow (E_\pi + m_p c^2)^2 - p_\pi^2 c^2 = (m_\Xi c^2 + m_{K^0} c^2 + m_{K^+} c^2)^2 \rightarrow$$

$$\rightarrow m_\pi^2 c^4 + \cancel{p_\pi^2 c^2} + m_p^2 c^4 + 2 m_p E_\pi c^2 - p_\pi^2 c^2 = (\ldots)^2 \rightarrow$$

$$\rightarrow E_\pi = \frac{(m_\Xi c^2 + m_{K^0} c^2 + m_{K^+} c^2) - m_\pi^2 c^4 - m_p^2 c^4}{2 m_p c^2}$$

$$E_i = m_i \gamma c^2$$

$$E_i = E_\pi + m_p c^2 = \gamma (m_\Xi c^2 + m_{K^0} c^2 + m_{K^+} c^2)$$

$$E_\Xi = m_\Xi \gamma c^2$$

DOMANDE TEORIA

- $\Delta t > 0$, $\Delta t' < 0$?

  SOLO space-like

- paradosso gemelli → torna più giovane

- $F = \underline{m\gamma a} = m\gamma \frac{v^2}{R} = qvB$

  $\omega = vR$

- $F^0 = \frac{\gamma}{c} \frac{dE}{dt}$

  $\underline{F} = m_0 \underline{A} = m_0 \frac{d\underline{U}}{d\tau}$

- $J^\mu = (\rho c, \vec{J})$

- $\rho' = -\gamma \frac{v}{c^2} J_x$